8 Read the passage below and answer the questions that follow.

Breakthrough Starshot: Laser-Driven Interstellar Probes

Breakthrough Starshot is a proposed space exploration project that aims to send extremely small spacecraft, known as Starchips, to the Alpha Centauri star system. Unlike conventional spacecraft that carry large amounts of fuel, Starshot intends to use a powerful Earth-based array of lasers to accelerate a lightweight spacecraft attached to a reflective light sail, as shown in Fig. 8.1. The concept relies on radiation pressure exerted by photons from the laser beam. Although photons have no rest mass, they carry momentum, and when they are reflected from a surface, they transfer momentum to that surface, producing a force.



Fig. 8.1

The nearest star system, Alpha Centauri, is approximately 4.37 light-years from Earth. Conventional spacecraft would require tens of thousands of years to travel this distance. The Starshot proposal aims to accelerate a spacecraft to approximately 0.20c, where c is the speed of light. At this speed, the journey could potentially be completed in around 20 years.

A major component of the project is the laser array. The proposed array consists of numerous synchronized laser emitters operating together as a phased array. By controlling the phase of each emitter, the individual beams combine into a highly focused beam directed toward the sail, delivering a total power of 100 GW at the sail.

The radiation force on the sail depends on the laser intensity and the sail reflectivity. A higher power output increases the force and therefore the acceleration of the spacecraft. The array must maintain accurate beam collimation. As the spacecraft moves further from Earth, beam divergence can cause the laser spot to spread, reducing the intensity incident on the sail. A larger array aperture reduces diffraction and helps maintain beam focus over long distances.

Atmospheric effects also influence the design. Turbulence in the Earth's atmosphere can distort the laser beam, reducing efficiency. Special adaptive optics systems may be required to compensate for atmospheric distortion. In addition, the array must dissipate heat generated by the laser equipment and maintain precise alignment among thousands of individual laser elements. The energy consumption of

the laser facility is expected to be extremely high, requiring substantial power generation and storage infrastructure.
One of the critical components of the spacecraft is the light sail. The sail must be extremely thin and lightweight to maximise acceleration according to Newton's second law. At the same time, it must possess high reflectivity so that most of the incident laser energy is reflected rather than absorbed. Reflection provides momentum transfer while minimising heating.

Material selection for the sail is therefore crucial. A highly reflective dielectric material may have a reflectivity exceeding 99.9% at the operating laser wavelength of 1.0 ⌈m. Nevertheless, a small fraction of the laser energy will always be absorbed. If the absorbed energy is not sufficiently removed, the sail temperature may rise to the point where structural failure occurs.

The rate of thermal energy absorbed P_{abs} by the sail can be calculated using

$$P_{abs} = (1\ R)P_{laser}$$

where *R* is the reflectivity and P_{laser} is the incident laser power.

To prevent overheating, the sail radiates thermal energy into space. Assuming the sail behaves approximately as a blackbody radiator with emissivity ε, the power radiated P_{rad} by the sail is described by the Stefan-Boltzmann law,

$$P_{rad} = \varepsilon\sigma AT^4$$

where

- $\sigma = 5.67 \times 10^{-8}\ W\ m^{-2}\ K^{-4}$ is the Stefan-Boltzmann constant,
- A is the total radiating area,
- T is the thermodynamic temperature of the sail,

If both sides of the sail radiate effectively, the total radiating area is twice the geometric area of the sail.

At thermal equilibrium,

$$P_{abs} = P_{rad}$$

This condition allows engineers to determine the operating temperature of the sail and assess whether the chosen material can survive the acceleration phase.

The sail must also remain mechanically stable. Uneven heating could produce thermal stresses that deform the sail and alter the direction of reflected laser light. Such deformation might reduce acceleration efficiency or even destabilise the spacecraft. Engineers must therefore optimise the sail thickness, shape, material properties and thermal conductivity.

The acceleration phase lasts only a few minutes. During this period, the spacecraft experiences extremely large accelerations. The onboard electronics must survive these forces while maintaining minimal mass. Since any increase in mass decreases acceleration for a given radiation force, all components must be carefully miniaturised.

A miniature power source is needed to power the camera, sensors and communication laser. No specific flight-ready power source has been identified. The mission concept assumes an ultra-lightweight power system of a few milligrams, likely involving advanced micro-batteries, radioisotope power, or another future technology that has not yet been fully developed.

(a) The laser beam hits the light sail at 90° and is totally reflected.

Starting from Newton's second law, show that the force exerted by the beam on the light sail is about 670 N.

[3]

(b) The spacecraft and sail have a total mass of 4.0 g.

(i) Determine the acceleration of the spacecraft.

acceleration = ... $m\ s^{-2}$ [1]

(ii) Determine the time taken for the spacecraft to accelerate to 0.20*c* from an initial speed of zero.

time = ... s [1]

(c) The spacecraft would have travelled approximately 1.0×10^{10} m during the acceleration phase.

The sail is circular and has a diameter of 5.0 m.

(i) Show that the angular width of the sail from Earth at the end of the acceleration phase is 5.0×10^{-10} rad.

[1]

(ii) Explain why it is not desirable for the beam width of the laser to be significantly greater than 5.0×10^{-10} rad.

..... [1]

(iii) If the laser beam comes from a single laser source, determine the diameter of the aperture of the laser source needed to restrict the beam width to 5.0×10^{-10} rad.

diameter = ... m [1]

(d) The light sail reflects 99.9% of the laser light incident on it. The emissivity ε is 0.80.

Determine the equilibrium temperature of the sail.

temperature = ... K [3]

(e) Suggest a reason why we do not consider heat loss through conduction and convection.

.....

..... [1]

(f) **(i)** The angular width of the sail from Earth is 1.0×10^{-6} rad at the start of the acceleration phase.

With reference to the answers to **(c)**, explain why the laser beam produces an uneven heating across the surface of the sail.

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..... [1]

(ii) Explain how the uneven heating across the surface of the sail can be reduced throughout the entire acceleration phase.

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..... [2]

(g) One of the options for an onboard power source is plutonium-238. Each plutonium-238 nucleus emits one α-particle of energy 5.48 MeV, of which only 6.7% can be harnessed as electrical energy.

The half-life of plutonium-238 is 86.4 years or 2.72×10^9 s.

The spacecraft requires at least 5.0 mW of power.

Determine the initial mass of plutonium-238 needed such that it can still supply this amount of power after 30 years.

mass = ... mg [5]

[Total: 20]